

Tutrugbu partial ATR (dis)harmony in Search

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Search, identifying the presence of a given configuration, can indirectly check if a string-peripheral segment is of a particular type: when, say, every high vowel from itself looks for the first vowel to its left, only the string-first high vowel, if there exists one, cannot find a preceding vowel. With this property, an analysis is proposed for the otherwise systematic ATR harmony between prefixes and a stem in Tutrugbu. The proposed analysis then gets adapted to incorporate the round harmony among some prefixes in the same language.

keywords: Tutrugbu; harmony; advanced tongue root (ATR); round (rounded, rounding, labial); Search

1. The phenomenon

In Tutrugbu (a Ghana-Togo Mountain language of Niger-Congo [Heine 2017: 277]), the ATR value of the prefix vowel(s) is determined by that of the first stem vowel (Essegbey 2009: 39, Essegbey 2019: 32):¹

(1) simplex ATR harmony (McCollum et al. 2020: 219, 220; tones omitted hereafter)²

- | | | | |
|----|----------------------|----|--------------------------------|
| a. | ɔ-da ‘CL3-copper’ | b. | ka-tɪ-ba-ba ‘CL7-NEG-FUT-come’ |
| | o-pete ‘CL3-vulture’ | | ke-ti-be-ʃe ‘CL7-NEG-FUT-grow’ |

When the respective two items in (1a) and (1b) are contrasted with each other, it is sure that the prefixes alternate. Since the first stem vowel is –ATR for the upper item in both columns, the prefix vowels are realized as –ATR. The +ATR realization happens for the same reason in the lower part.³ This straightforwardness of the ATR agreement gets blurred if *both* i) the first vowel of a given string is a high vowel *and* ii) there exists a prefixal non-high vowel (McCollum et al. 2020: 220, 221): it is the –ATR-exponence of the alternating prefixes that co-occurs with a stem whose first vowel is +ATR, as shown in the first two examples below. Between the parentheses, what is squared is meant to show that these first-met affixes indeed alternate.

(2) disharmony under the specified conditions (McCollum et al. 2020: 219, 220, 221)

¹ The emphasis on the first stem vowel is due to the existence of disharmonic stems like hɔlitsa ‘chameleon’ (Essegbey 2019: 35).

² On the surface, ɪ and ʊ are realized as ɛ and ɔ, respectively (McCollum et al. 2020: 218, McCollum and Essegbey 2020: 3, 27).

³ ɛ does not occur in “prefixes” (McCollum and Essegbey 2020: 3) / “affixes” (McCollum et al. 2020: 218).

- a. bu-ba-fe ‘1PL-FUT-grow’ (cf. bu-ti-fe ‘1PL-NEG-grow’)
 b. i-ti-ka-a-ba-ba-wu ‘1SG-NEG-PFV-PROG-VENTIVE-VENTIVE-climb’
 (cf. i-ti-wu ‘1SG-NEG-climb’
 e-ti-ke-e-be-be-wu ‘3SG-NEG-PFV-PROG-VENTIVE-VENTIVE-climb’)
 c. bu-ba-di-wu ‘1PL-FUT-ITIVE-climb’ (cf. i-di-to ‘1SG-ITIVE-cook’)

The disharmony, nonetheless, is partial. When the aforementioned conditions are met but there are prefixal high vowels which follow the rightmost non-high vowel, the ATR harmony happens as expected. For the left half of 1PL-FUT in the last example above, the disharmony conditions are met and the –ATR exponents occur; as for the right half, the itive prefix vowel is +ATR due to the verb’s +ATR vowel.

2. Underlying representation

For the analysis to come, I will assume that the relevant features are Syllabic, High, Low, Back, Round, and ATR. All the non-alternating vowels in affixes or not are fully specified for these six features. As for the alternations $i \sim \text{ɪ}$, $u \sim \text{ʊ}$, $o \sim \text{ɔ}$, and $e \sim \text{ə}$, they have the following representations after set intersection:

- I: {+syllabic, +high, –low, –back, –round}
 i: {+syllabic, +high, –low, –back, –round, ≠ATR}
 ɪ: {+syllabic, +high, –low, –back, –round, =ATR}
 U: {+syllabic, +high, –low, +back, +round}
 u: {+syllabic, +high, –low, +back, +round, ≠ATR}
 ʊ: {+syllabic, +high, –low, +back, +round, =ATR}
 O: {+syllabic, –high, –low, +back, +round}
 o: {+syllabic, –high, –low, +back, +round, ≠ATR}
 ɔ: {+syllabic, –high, –low, +back, +round, =ATR}
 Æ: {+syllabic, –high, –round}
 e: {+syllabic, –high, =low, =back, –round, ≠ATR}
 ə: {+syllabic, –high, ≠low, ≠back, –round, =ATR}

The crossed valued features are left out in the uppercase representations because they are different from each other in the indented pairs.

3. Search and Change

The rules to be presented are composed of (substance-free) Search and Change (Dabbous et al.

2022). Search (Mailhot and Reiss 2007) identifies if a given configuration is present. It consists of three parameters: i) the initiators (INRs) are *all* the segments whose feature bundles are supersets of the set $\{\pm F_1, \pm F_2 \dots\}$; ii) the direction (DIR) instructs Search to proceed either leftward or rightward from an INR; iii) the terminator (TRM) is the *first* segment whose feature bundle is a superset of the set $\{\pm G_1, \pm G_2 \dots\}$ to the designated side of an INR. This operation of seeking stops when either the TRM is found or one of the two end domains is reached, which means that Search can return empty-handed.

Change has two intrasegmental operations: subtraction (−) and unification ($\sqcup$). Subtraction (Bale et al. 2014: 246) is set difference and defined as:

$$A - B = \{aX: aX \in A \wedge aX \notin B\}$$

If, say, +high is removed from a high (i.e. +high) vowel, the resulted vowel is High-unspecified. The same removal, when applied to a(n already) High-unspecified vowel and a non-high (i.e. −high) one, shows no change since no +high exists. And these vowels remain High-unspecified and non-high after this removal. Unification (ibid.: 245) is defined as:

$$A \sqcup B = \begin{cases} A \cup B, & \text{if } \forall F_i [(F_i \in \{F_1, \dots, F_n\}) \rightarrow (\{+F_i, -F_i\} \not\subseteq A \cup B)]; \\ A, & \text{otherwise.} \end{cases}$$

It is like union if the result does not contain for any features of the universal features set $(\{F_1, \dots, F_n\})$ both the plus-valued version and the minus-valued one. For any other cases, unification shows no change for the undergoers. For instance, when a −ATR vowel is unified with +ATR, the vowel is still −ATR afterwards—an instance of unification failure. As for a +ATR vowel, the unification with +ATR vacuously results in the vowel with +ATR specification. When +ATR unification takes place with an ATR-unspecified vowel, a +ATR vowel is derived non-vacuously.

When used in the context of Change, unaffixed INR is to denote the initiators of a given search with terminator and to name the set which includes all the successful initiators of a given search. In the same context, complement-notation prefixed CINR is to denote the initiators of a given search without terminator and to name the set which includes all the failed initiators of a given search.

4. Three ordered rules

To account for the conditioned disharmony, the rule below is applied first:

(3) −ATR-insertion

Search INR-i: [+syllabic, –high]

INR-j: [+syllabic, +high, –low]; DIR-j: left; TRM-j: [+syllabic]

Change INR-i $\sqcup$ {–ATR}, if $[\exists x (x \in \text{CINR-j})]$

‘Unify a non-high vowel with –ATR, if there exists a high vowel which looks leftward for a vowel but does not find it.’

Notice the difference between brackets and braces (Bale et al. 2014: 243)—of course, this distinction does not apply to predicate logic. Braces, as usual, are employed to demark sets. Brackets [...] are used to abbreviate $\{x: x \supseteq \{\dots\}\}$. So, [+syllabic, –high] refers to the natural class of non-high vowels since the set {+syllabic, –high} is a subset of every non-high vowel; in other words, every non-high vowel is a superset of the set {+syllabic, –high}. The co-occurrence of +high and –low in [+syllabic, +high, –low] may need some comments. Based on Hale and Reiss (2008: 96-97, 100, 274-275), what is specified for INR and TRM is the product of set intersection across the segments involved. Here, the valued features shared among the high vowels are +syllabic, +high, and –low, so the natural class is displayed as such. All the rules in the current study follow this practice.

Now look more into Search-j and its result as the condition on the unification of –ATR with any non-high vowels. By letting each high vowel look leftward for another vowel, the purpose is to check whether the string-first vowel is high or not. And the three possible scenarios are described below:

scenario i—there are no high vowels

Since there are no initiators, it is meaningless to talk about success/failure. The set CINR-j is empty, so the if-condition is not met.

scenario ii—all the high vowels are the second or any of the following vowels of the string
Each initiator of Search-j is always preceded by another vowel and successfully finds its terminator. The set CINR-j is thus empty, so the if-condition is not met.

scenario iii—the first vowel of the string is a high vowel

Due to the unsuccessful search initiated by the string-first high vowel, the set CINR-j is nonempty. Hence, the if-condition is met, so all the non-high vowels are unified with –ATR.

Next comes the ATR agreement:

(4) ATR agreement

Search INR: [+syllabic]; DIR: right; TRM: [+syllabic, α ATR]

Change INR $\sqcup$ { α ATR}

‘Unify a vowel with the ATR value of the first ATR-specified vowel to its right.’

I practice here the identity use of Greek letter variables (Bale and Reiss 2018: 451 vs. 454-455), and my understanding is that there are at least two occurrences of a given variable. Moreover, I assume that Search and Change must each host at least one such occurrence for a specific variable used. I take these two moves, for now, as my working hypothesis.

Up to this point, I, U, and O are no longer present due to the application of the rules (3) and (4) (in that order). Nonetheless, $\mathcal{A}$ E has only become either A: {+syllabic, –high, –round, –ATR} or E: {+syllabic, –high, –round, +ATR}. In order to complete the current analysis, a context-free rule is needed, where the parameters DIR and TRM are left blank and not mentioned:

(5) exchange addition

Search INR: [+syllabic, –high, –round, α ATR]

Change INR $\sqcup$ {– α low} $\sqcup$ {– α back}

‘Unify an ATR-specified non-high unrounded vowel with the opposite value to that of ATR for the features Low and Back.’

Think of the two tokens of the unification symbol as the two tokens of the plus sign in, say, “y+17+29”. INR is unified with {– α low} and becomes INR', and INR' is unified with {– α back} and becomes INR"—as one is doing addition multiple times. This last rule, for the needed ATR value is cumulated from the earlier two, changes A and E into a and e, respectively.

To make concrete the foregoing discussion, the derivation of bu-ba-di-wu ‘1PL-FUT-ITIVE-climb’ from (2c) is given:

(6) an annotated derivation

underlying form: bU-b $\mathcal{A}$ -dI-wu

–ATR-insertion

Search	INR-i: [+syllabic, –high]— $\mathcal{A}$ E		
	INR-j: [+syllabic, +high, –low]		TRM-j: [+syllabic]
	u	leftward	I
	I	finds	$\mathcal{A}$ E
	U		nothing
Change	the set C[INR-j: {U}] is a singleton , so INR-i $\sqcup$ {–ATR}		
	$\mathcal{A}$ E becomes A		

intermediate form: bU-bA-dI-wu

ATR agreement

Search	INR: [+syllabic]		TRM: [+syllabic, α ATR]
	U		A: {...-ATR...}
	A	rightward	u: {...+ATR...}
	I	finds	u: {...+ATR...}
	u		nothing

Change INR $\sqcup$ { α ATR}
 U becomes υ
 A does not change because of unification failure
 I becomes i
 u does not change because of no terminator

intermediate form: b υ -bA-di-wu

exchange addition

Search	INR: [+syllabic, -high, -round, α ATR]—A: {...-ATR...}
Change	INR $\sqcup$ {- α low} $\sqcup$ {- α back}
	A becomes a: {...+low, +back...}

surface form: b υ -ba-di-wu

5. Rounding harmony

According to McCollum and Essegbey (2020), there exists a four-way alternation $e \sim o \sim a \sim \text{ɔ}$ (7a-d) for some prefixes in Tutrugbu, as shown in (7) for the future prefix.⁴

(7) rounding harmony (McCollum et al. 2020: 219; McCollum and Essegbey 2020: 5-6, in the former's transcription system)

- | | |
|--------------------------------------|----------------------------------|
| a. e-be-fe '3SG-FUT-grow' | b. o-bo-fe '2SG-FUT-grow' |
| c. a-ba-ba '3SG-FUT-come' | d. ɔ-bɔ-ba '2SG-FUT-come' |
| e. no-ti-bo-wu '2P-NEG-FUT-climb' | f. nɔ-tɪ-bɔ-ba '2P-NEG-FUT-come' |
| g. b υ -ba-ti '1P-FUT-know' | |

The ATR-distinction between $a \sim \text{ɔ}$ and $e \sim o$ was discussed in the earlier paragraphs. The round-distinction between $a \sim e$ and $\text{ɔ} \sim o$ is due to the mid rounded vowel contained within some farther preceding prefixes; in (7g), the high vowel of the first person plural prefix is rounded but does not change the non-high vowel of the future prefix into its rounded counterpart. It is shown in

⁴ The earlier $e \sim a$ alternation is subsumed under this alternation.

(7e-f) that the non-high vowels involved need not be within the two adjacent syllables. I speculate below how this alternation could be incorporated into what has been suggested.

To begin with, the underlying representation for this four-way allomorphy is $\{+syllabic, -high\}$ by means of set intersection among these four segments. The rounded realization is attributed to:

(8) W-coloring

Search INR: $[+syllabic, -high]$; DIR: left; TRM: $[+syllabic, -high, -low, +back, +round]$

Change INR $\sqcup \{+round\} \sqcup \{+back\} \sqcup \{-low\}$

‘Unify +round, +back, and -low with a non-high vowel which is preceded by a mid back rounded vowel.’

Recall that Search may or may not succeed, so, after the application of (8), the roundedness may still remain unspecified. For this and other lack of specifications, I will modify (5) into:

(5') exchange plus

Search INR: $[+syllabic, -high, \alpha ATR]$

Change INR $\sqcup \{-\alpha low\} \sqcup \{-\alpha back\} \sqcup \{-round\}$

‘Unify an ATR-specified non-high vowel with -round and the opposite value to that of ATR for the features Low and Back.’

The aim for +round to surface sometimes forces (8) to transpire before (5').

As a wrap-up, how to derive the $o \sim \text{ɔ} \sim e \sim a$ alternation is demonstrated. Although the ascertainable rule ordering is: $\left. \begin{matrix} (3) < (4) \\ (8) \end{matrix} \right\} < (5')$, the demonstration, for ease of typography, starts with (8).

(9) schematic derivation of the four-way alternation

input	$\{+syllabic, -high\}$			
W-coloring	$\left\{ \begin{matrix} +syllabic \\ -high \\ +round \\ +back \\ -low \end{matrix} \right\}$	$\left\{ \begin{matrix} +syllabic \\ -high \\ +round \\ +back \\ -low \end{matrix} \right\}$	<i>search failure and no change</i>	
-ATR-insertion ATR agreement	$\left\{ \begin{matrix} +syllabic \\ -high \\ +round \\ +back \\ -low \\ +ATR \end{matrix} \right\}$	$\left\{ \begin{matrix} +syllabic \\ -high \\ +round \\ +back \\ -low \\ -ATR \end{matrix} \right\}$	$\left\{ \begin{matrix} +syllabic \\ -high \\ +ATR \end{matrix} \right\}$	$\left\{ \begin{matrix} +syllabic \\ -high \\ -ATR \end{matrix} \right\}$

exchange plus	<i>unification failure and/ or vacuous unification</i>				$\left\{ \begin{array}{l} +\text{syllabic} \\ -\text{high} \\ +\text{ATR} \\ -\text{round} \\ -\text{low} \\ -\text{back} \end{array} \right\}$	$\left\{ \begin{array}{l} +\text{syllabic} \\ -\text{high} \\ -\text{ATR} \\ -\text{round} \\ +\text{low} \\ +\text{back} \end{array} \right\}$
output	o	ɔ	e	a		

In the row pertinent to ATR, no differentiation is made for the sources of –ATR: due to the string-first vowel being high or because of a stem’s first vowel being –ATR. The source of +ATR, nonetheless, can only be the +ATR of the stem-first vowel.

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